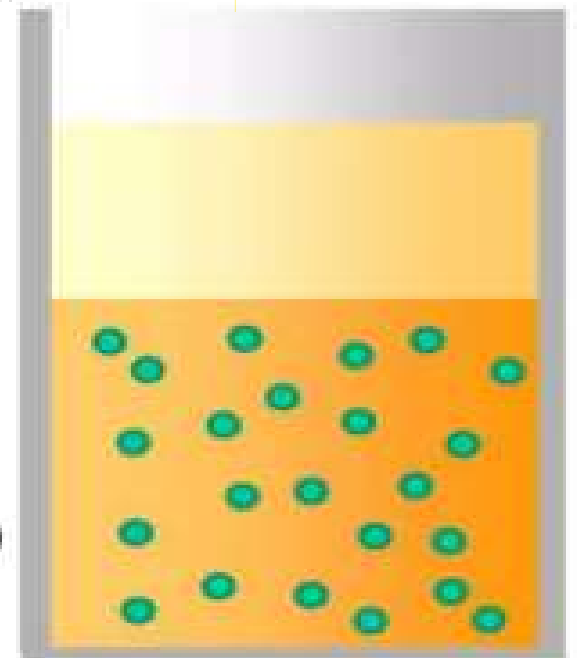


COLLOIDS

- A colloid is a substance microscopically dispersed throughout another substance
- The word colloid comes from a Greek word '**kolla**', which means glue thus colloidal particles are glue like substances.
- These particles pass through a filter paper but not through a semipermeable membrane.
- Colloids can be made settle by the process of centrifugation.

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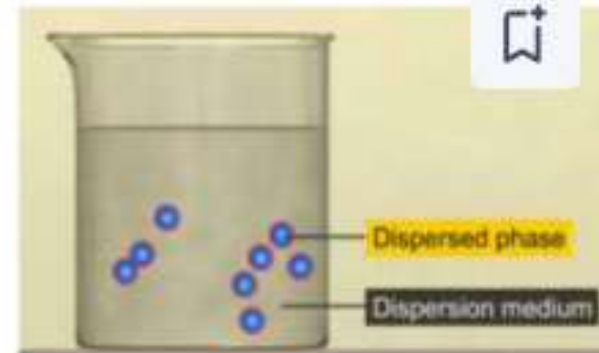


- Colloids can be made settle by the process of centrifugation.

- The colloidal system consist of two phases:

A dispersed phase (A discontinuous phase)

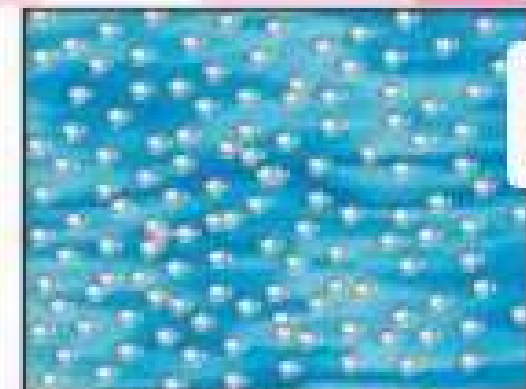
A dispersion medium (A continuous phase)



- The dispersed-phase particles have a diameter of between approximately **1nm – 100nm**.
- Such particles are normally invisible in an optical, though their presence can be confirmed with the use of an **ultramicroscope or an electron microscope**.

SOLUTIONS

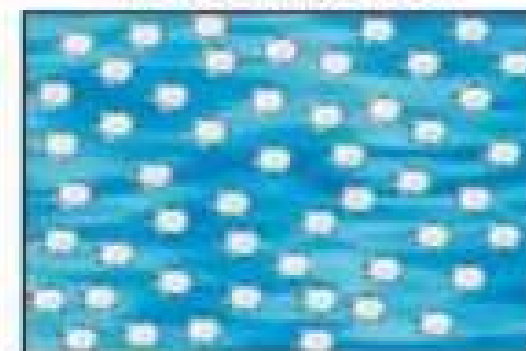
- ❑ Made up of particles or solutes and a solvent
- ❑ The solvent part of the solution is usually a liquid, but can be a gas.
- ❑ The particles are atoms, ions, or molecules that are **very small in diameter**.



True solution

COLLOIDAL MIXTURE

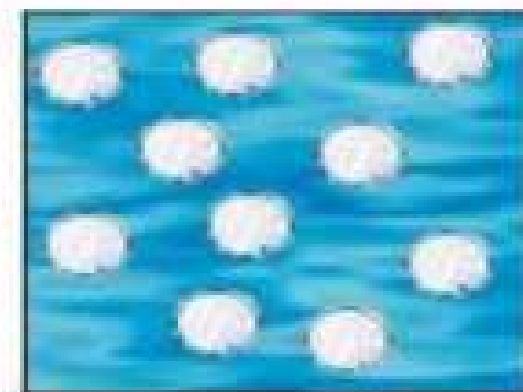
- ❑ Has particles that are not as small as a solution and not as large as a suspension.
- ❑ The particles **are intermediate in size**.



Colloidal solution

SUSPENSIONS

- ❑ Made up of particles and a solvent
its **particles are larger** than those found in a solution.
- ❑ The particles in a suspension can be distributed throughout the suspension evenly by shaking the mixture.



Suspensions



Solution:

Table salt dissolves in water to form Salt (saline) water. A solute (salt; NaCl) is dissolved in another substance (water) known as a solvent, and this creates a solution.



Suspension:

Flour suspended in water (appears light blue because blue light is scattered off the flour particles to a greater extent than red light)



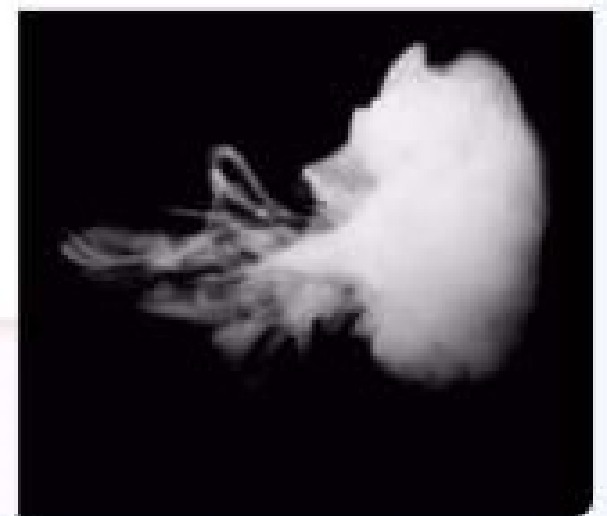
Colloid:

Milk is an emulsified colloid of liquid butterfat globules dispersed within a water based liquid. Colloids are Stabilized In suspension by Electrostatics - mutual Repulsion of like electrical Charges.

Comparison of the Properties of Solutions, Colloids, And Suspensions

Property	True Solution	Colloid	Suspension
Particle Size	Less than 1 nm	1 to 100 nm	More than 100 nm
Appearance	Clear	Cloudy	Cloudy
Homogeneity	Homogeneous	Homogeneous or Heterogeneous	Heterogeneous
Transparency	Transparent but often coloured	Often translucent and opaque but can be transparent	Often opaque but can be translucent
Separation	Does not separate	Can be separated	Separates or settles
Filterability	Passes through filter paper	Passes through filter paper	Particles do not pass through filter paper

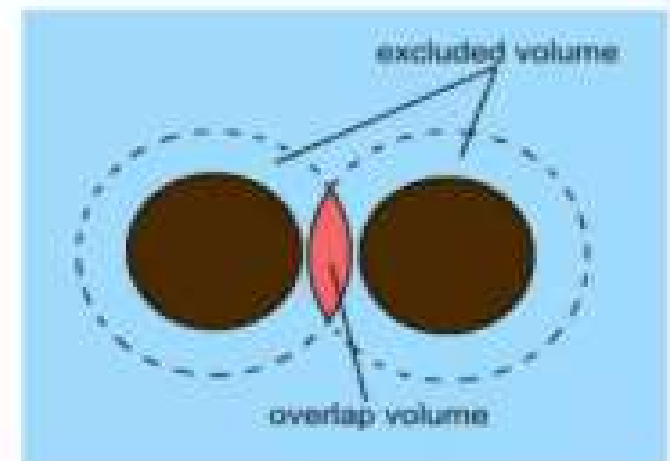
➤ Examples of colloids are milk, synthetic polymers, fog, blood, jam, shoe polish, smoke, etc.



The following forces play an important role in the interaction of colloid particles:

EXCLUDED VOLUME REPULSION IN LIQUID THEORY :

In liquid state theory, the 'excluded volume' of a molecule is the volume that is **inaccessible to other molecules** in the system as a result of the presence of the first molecule.



The excluded volume of a hard sphere is eight times its

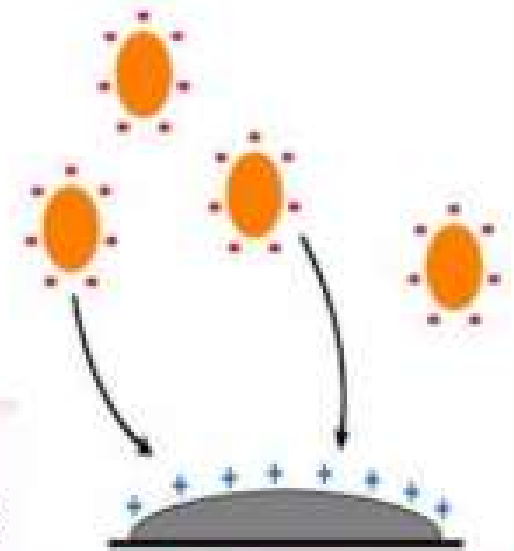
It is the sum of the attractive or repulsive forces between molecules other than those due to covalent bonds, the hydrogen bonds, or the electrostatic interaction of ions with one another



ELECTROSTATIC INTERACTION : or with neutral molecules or charged molecules.

Colloidal particles often carry an electrical charge and therefore attract or repel each other.

The charges of both the continuous and the dispersed phase, as well as the mobility of the phases are factors affecting

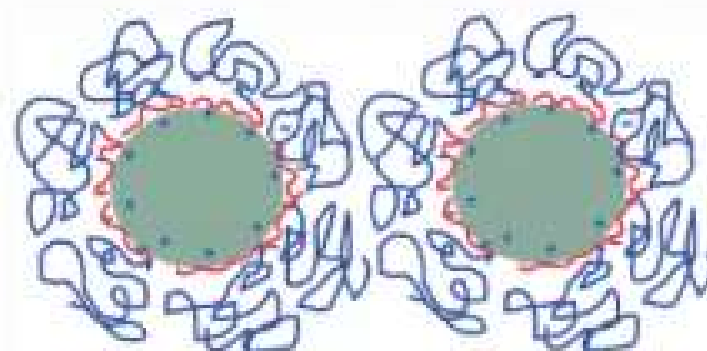


STERIC



Steric effects arise from the fact that each atom within a molecule occupies **a certain amount of space**.

If atoms are brought too close together, there is an associated cost in energy due to overlapping electron clouds (Pauli or Born repulsion), and this may affect the molecule's preferred shape (conformation) and reactivity.



CLASSIFICATION OF COLLOIDS

- ☐ Based of physical state of dispersed phase an dispersion medium.
- ☐ Based of nature of interaction between dispersed phase and dispersion medium.
- ☐ Based on molecular size in the dispersed phase.
- ☐ Based on appearance of colloids.
- ☐ Based on electric charge on dispersion phase.

Based on physical state of dispersed phase and dispersion medium

Dispersed Phase	Dispersion Medium	Name	Examples
Solid	Solid	Solid-Sol	Alloys, Cranberry glass
Solid	Liquid	Sol	Ink, Blood
Solid	Gas	Aerosol	Smoke, Ice cloud
Liquid	Solid	Gel	Jelly, Curd
Liquid	Liquid	Emulsion	Milk, Cream
Liquid	Gas	Liquid aerosol	Cloud, Fog
Gas	Solid	Solid form	Aerogel, Pumice stone
Gas	Liquid	Foam	Shaving cream
Gas	Gas	None	All gases are miscible

BASED ON NATURE OF INTERACTION BETWEEN DISPERSED PHASE AND DISPERSION MEDIUM



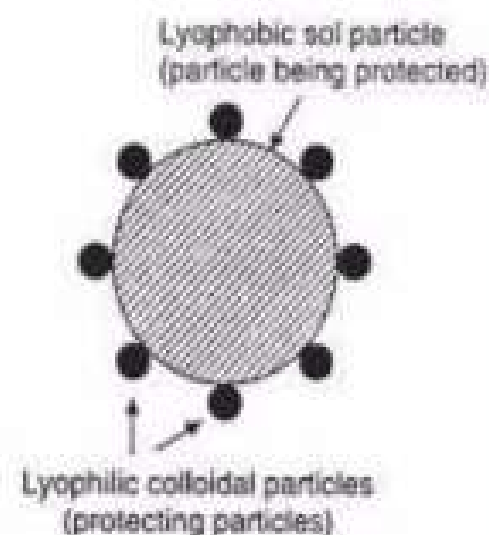
LYOPHILIC COLLOIDS

- Colloidal solution in which the dispersed phase has a great affinity for the dispersion medium.
- They are also termed as **intrinsic colloids**.
- Such substances have tendency to pass into colloidal solution when brought in contact with dispersion medium.
- If the dispersion medium is water, they are called **hydrophilic or emulsoids**.
- The lyophilic colloids are generally **self-stabilized**.
- **Reversible** in nature and are heavily hydrated.
- Example of lyophilic colloids are starch, gelatin, rubber, protein etc.



LYOPHOBIC COLLOIDS

- Colloidal solutions in which the dispersed phase has no affinity to the dispersion medium.
- These are also referred as **extrinsic colloids**.
- Such substances have no tendency to pass into colloidal solution when brought in contact with dispersion medium.
- The lyophobic colloids are relatively **unstable**.
- They are **irreversible** by nature and are stabilized by adding small amount of electrolyte.
- They are poorly hydrated.
- If the dispersion medium is water, the lyophobic colloids are called **hydrophobic or suspensoids**.
- Examples, sols of metals like Au, Ag, sols of metal hydroxides and sols of metal sulphides.



Based on molecular size in the dispersed phase.



➤ MULTIMOLECULAR COLLOIDS

Individual particles of the dispersed phase consists of **aggregates of atoms or small molecules** having diameter less than 10^{-7}cm . The particles are held by weak **vander waal's forces**.

Example; gold sol, sulphur sol

➤ MACROMOLECULAR COLLOIDS

The particles of dispersed phase are sufficiently **large in size** enough to be of colloidal solution. These are called Natural Polymers.



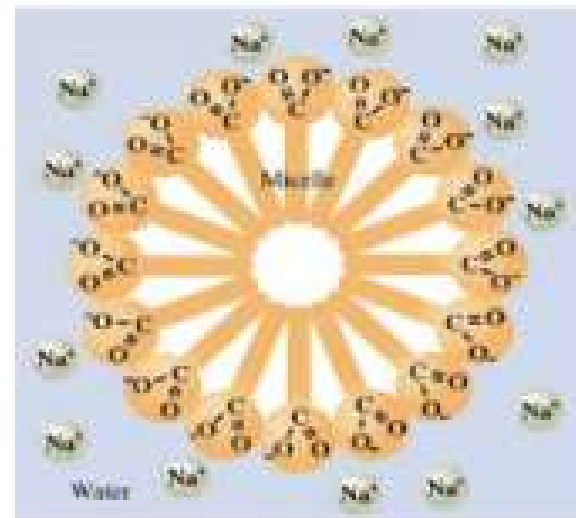
Associated Colloids

These colloids behave as normal electrolytes at low concentrations but behave as colloids at higher concentrations.

These associated colloids are also referred to as **micelles**.

Sodium stearate ($C_{18}H_{35}NaO_2$) behave as electrolyte in dilute solution but colloid in higher concentrations.

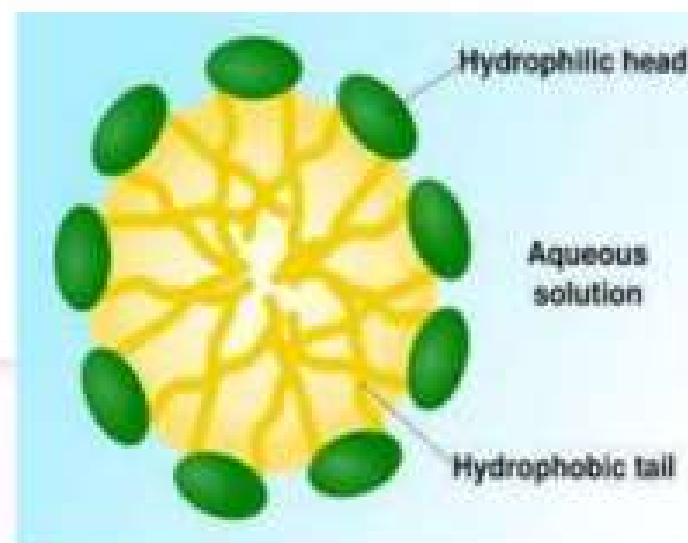
Examples: Soaps , higher alkyl sulphonates , polythene oxide.



HOW DOES SOAP WORK?



- When greasy dirt is mixed with soapy water, the soap molecules arrange themselves into tiny clusters called **micelles**.
- The water-loving (hydrophilic) part of the soap molecules sticks to the water and points outwards, forming the outer surface of the micelle.
- The oil-loving (hydrophobic) parts stick to the oil and trap oil in the center where it can't come into contact with the water. With the oil tucked safely in the center, the



Based on appearance of colloids



• SOLS

- When a colloidal solution appears as **fluid**.
- The sols are generally named as dispersion medium.
- When the dispersion medium is water, the sol is known as **hydrosol or aquosol**.
- When the dispersion medium is alcohol or benzene it is called **alcosol and benzosol** respectively.



• GELS

- When a colloidal solution appear as **solid**.
- The rigidity of gel varies from substance to substance.
- Examples : jelly, butter, cheese, curd.



BASED ON ELECTRICAL CHARGE ON DISPERSION PHASE



➤ POSITIVE COLLOIDS

When dispersed phase in a colloidal solution carries **a positive charge**.

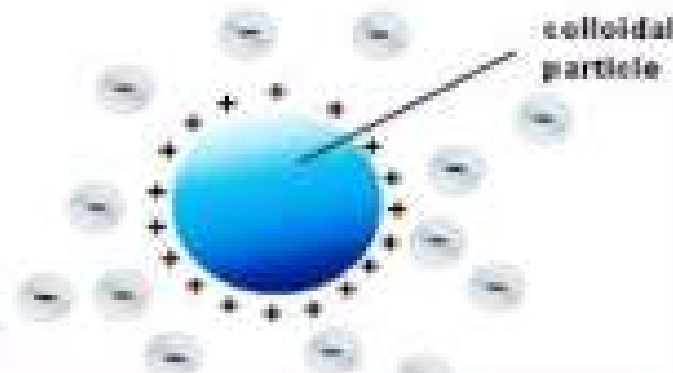
Examples : Metal hydroxides like $\text{Fe}(\text{OH})_3$, $\text{Al}(\text{OH})_3$, methylene blue sol etc.



➤ NEGATIVE COLLOIDS

When dispersed phase in a colloidal solution carries **a negative charge**.

Examples : Ag sol, Cu sol



Preparation of Colloids

❖ *Mechanical Dispersion*

❖ *Electrical dispersion or Bredig's Arc Method*

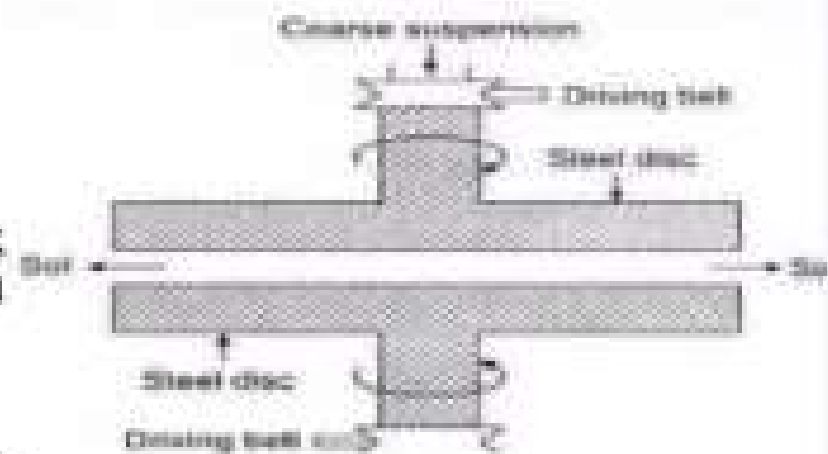
❖ *Peptisation*

❖ *Condensation Method*

A) Mechanical dispersion:

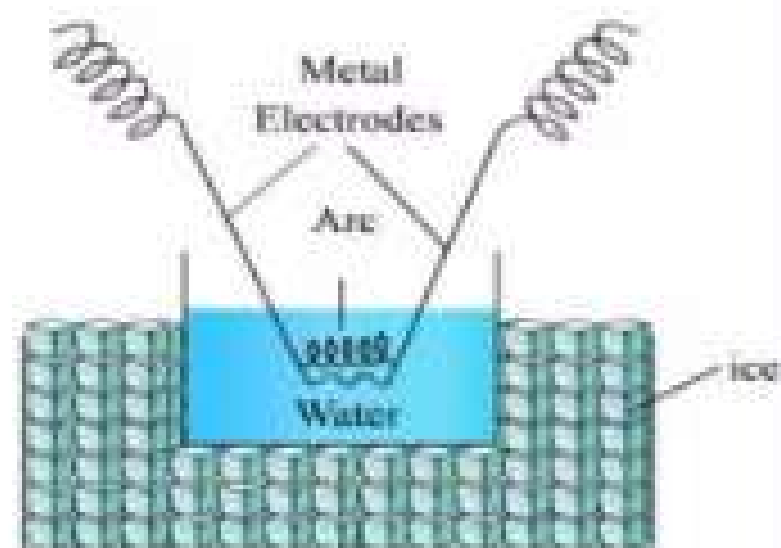
In this method,

- The substance is first ground to coarse particles.
- It is then mixed with the dispersion medium to get a suspension.
- The suspension is then grinded in colloidal mill.
- It consists of two metallic discs nearly touching each other and rotating in opposite directions at a very high speed about 7000 revolution per minute.
- The space between the discs of the mill is so adjusted that coarse suspension is subjected to great shearing force giving rise to particles of colloidal size.
- Colloidal solutions of black ink, paints, varnishes, dyes etc. are obtained by this method.



(B) By electrical dispersion or Bredig's arc method:

- This method is used to prepare sols of platinum, silver, copper or gold.
- The metal whose sol is to be prepared is made as two electrodes which immerse in dispersion medium such as water etc.
- The dispersion medium is kept cooled by ice.
- An electric arc is struck between the electrodes.
- The tremendous heat generated by this method give colloidal solution.
- The colloidal solution prepared is stabilized by adding a small amount of KOH to it.



Bredig's arc method

(C) By peptisation:

- The process of converting a freshly prepared precipitate into colloidal form by the addition of suitable electrolyte is called peptisation.
- Cause of peptisation is the adsorption of the ions of the electrolyte by the particles of the precipitate.
- The electrolyte used for this purpose is called peptizing agent or stabilizing agent.
- Important peptizing agents are sugar, gum, gelatin and electrolytes.



(D) Condensation method

- In condensation method, the smaller particles of the dispersed phase are aggregated to form larger particles of colloidal dimensions.
- Some important condensation methods are described below:
 - a) Solutions of substances like mercury and sulphur are prepared by passing their vapours through a cold water containing a suitable stabilizer such as ammonium salt or citrate.



d) Chemical methods:

Colloids can be prepared by following chemical methods.

1) Oxidation:

Addition of oxygen and removal of hydrogen is called oxidation.

For example, Colloidal solution of sulphur can be prepared by oxidizing an aqueous solution of H_2S with a suitable oxidizing agent such as bromine water.



2) Reduction:

Addition of hydrogen and removal of oxygen is called reduction.

For example: Gold sol can be obtained by reducing a dilute aqueous solution of gold with stannous chloride.



3) Hydrolysis:

It is the break down of water.

Sols of ferric hydroxide and aluminium hydroxide can be prepared by boiling the aqueous solution of the corresponding chlorides.

For example,



Purification of colloids

There are three common methods used for purification of colloids:

- Dialysis
- Electrodialysis
- Ultra filtration

Dialysis

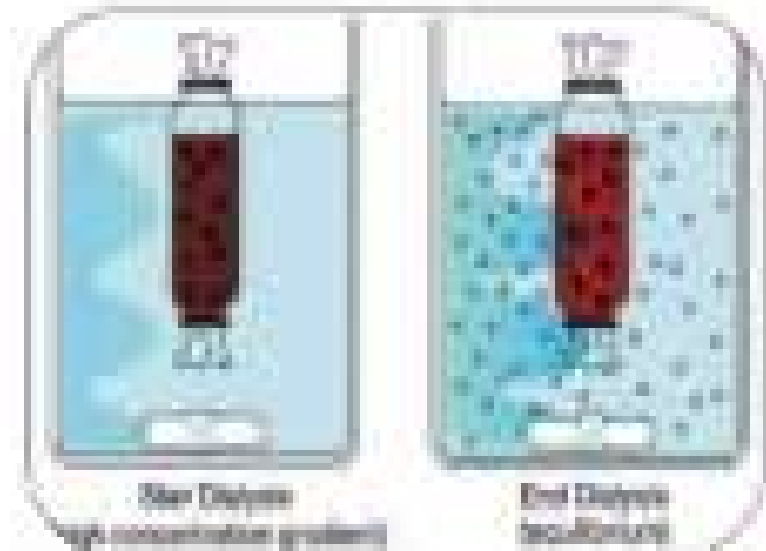
➤ The process of separating the particles of colloid from those of crystalline by means of diffusion through a suitable membrane.

➤ Its principle is that **colloidal particles can not pass through a cellophane membrane while the ions of the electrolyte can pass through it.**

➤ The impurities slowly diffused out of the bag leaving behind pure colloidal solution.

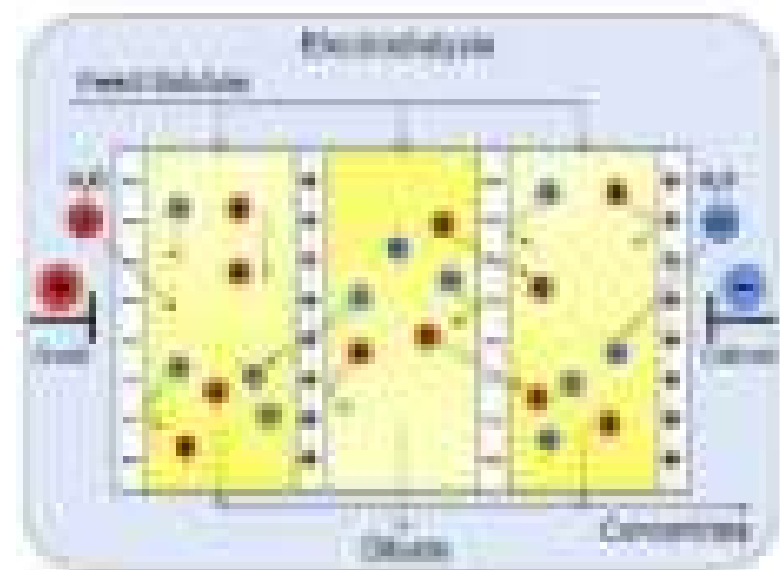
➤ The distilled water is changed frequently to avoid accumulation of the crystalline otherwise they may start diffusing back into the bag.

➤ Dialysis can be used for removing HCl from the ferric hydroxide sol.



Electro dialysis

- The process of dialysis is very slow.
- The process is speeded up by application of electrical potential.
- This is called electro dialysis.



Ultra filtration

- Ultra filtration is a process of high pressure

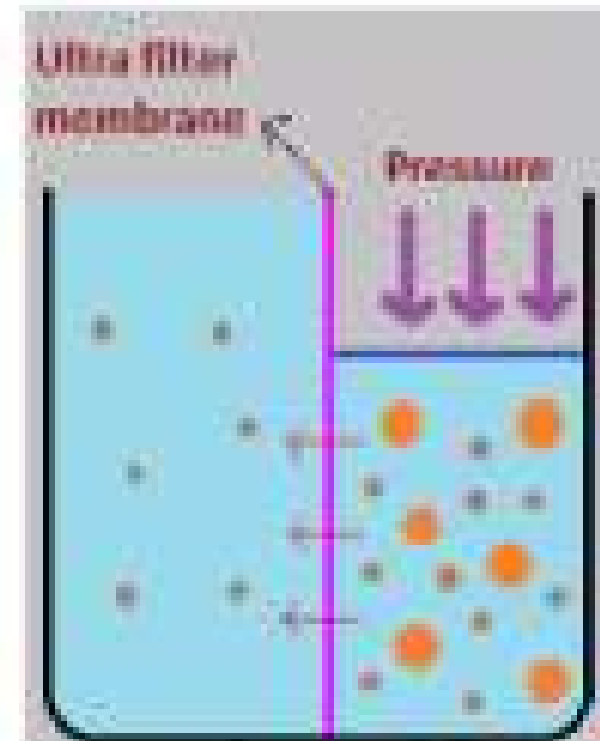
filtration through a semi permeable membrane

in which colloidal particles are retained while

the small sized solutes and the solvent are

forced to move across the membrane by

hydrostatic pressure forces.



PHYSICAL PROPERTIES OF COLLOIDS

- **Heterogeneity:** Colloidal solutions consist of two phases-dispersed phase and dispersion medium.
- **Visibility of dispersed particles:** The dispersed particles present in them are not visible to the naked eye and they appear homogenous.
- **Filterability:** The colloidal particles pass through an ordinary filter paper. However, they can be retained by animal membranes, cellophane membrane and ultrafilters.
- **Stability:** Lyophilic sols in general and lyophobic sols in the absence of substantial concentrations of electrolytes are quite stable.
- **Colour:** The colour of a colloidal solution depends upon the size of colloidal particles present in it. Larger particles absorb the light of longer wavelength and therefore transmit light of shorter wavelength.

OPTICAL PROPERTIES OF SOLS

(1) Sols exhibit Tyndall effect

When a strong beam of light is passed through a sol and viewed at right angles, the path of light shows up as a hazy beam or cone. This is due to the fact that sol particles absorb light energy and then emit it in all directions in space. This 'scattering of light', as it is called, illuminates the path of the beam in the colloidal dispersion.

The phenomenon of the scattering of light by the sol particles is called Tyndall effect.

The illuminated beam or cone formed by the scattering of light by the sol particles is often referred as **Tyndall beam** or **Tyndall cone**.

The hazy illumination of the light beam from the film projector in a smoke-filled theatre or the light beams from the headlights of car on a dusty road, are familiar examples of the Tyndall effect. If the sol particles are large enough, the sol may even appear turbid in ordinary light as a result of Tyndall scattering.

True solutions do not show Tyndall effect. Since ions or solute molecules are too small to scatter light, the beam of light passing through a true solution is not visible when viewed from the side. Thus Tyndall effect can be used to distinguish a colloidal solution from a true solution.

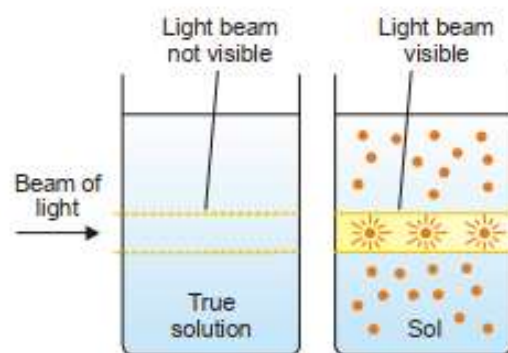


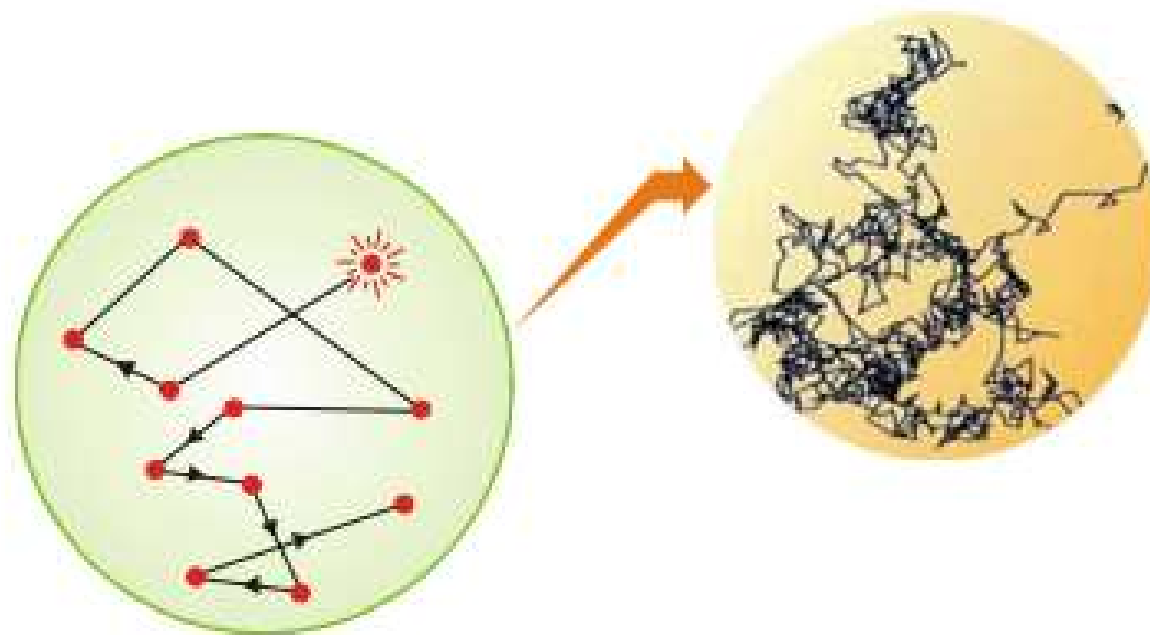
Figure 22.10
Tyndall effect (Illustration).

KINETIC PROPERTIES OF SOLS

Brownian Movement

When a sol is examined with an ultramicroscope, the suspended particles are seen as shining specks of light. By following an individual particle it is observed that the particle is undergoing a constant rapid motion. It moves in a series of short straight-line paths in the medium, changing directions abruptly.

The continuous rapid zig-zag movement executed by a colloidal particle in the dispersion medium is called **Brownian movement or motion**.



■ **Figure 22.13**

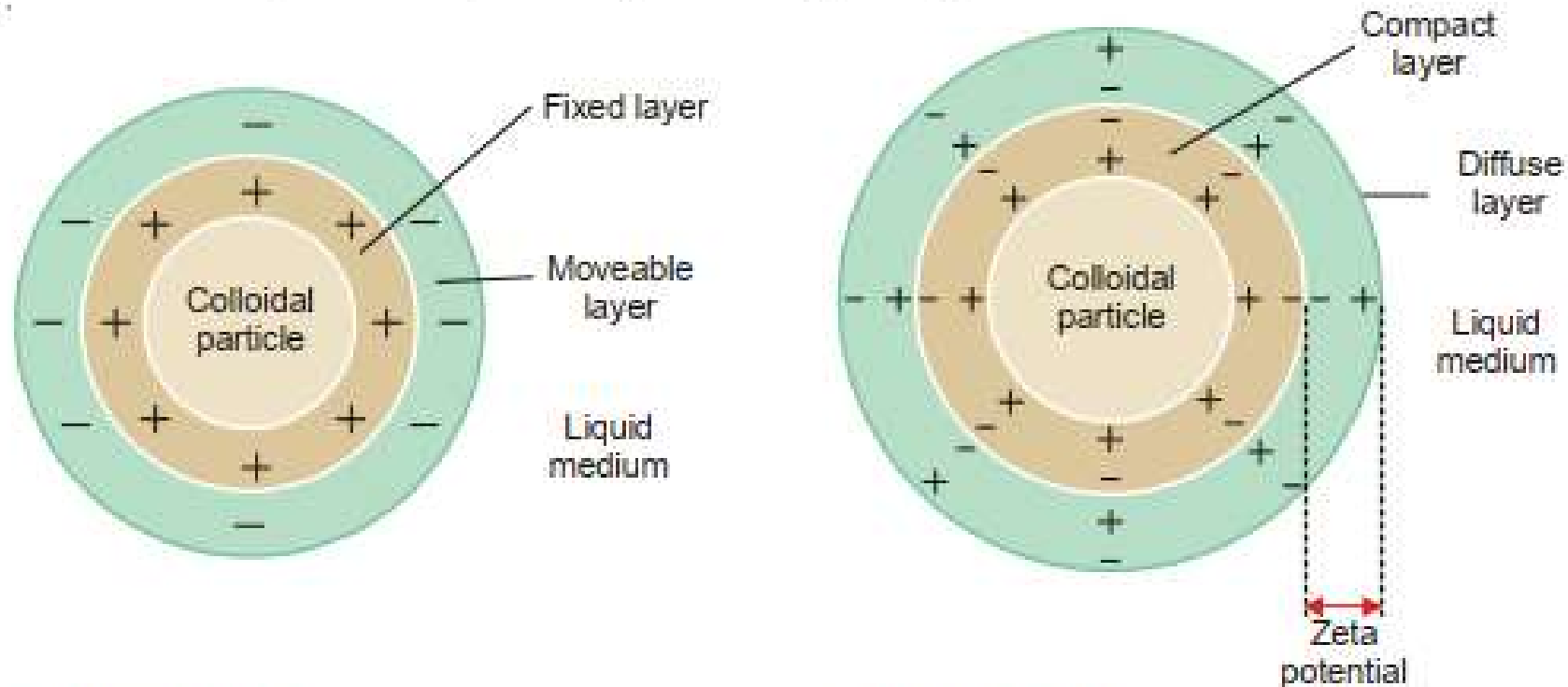
An illustration of Brownian movement.

This phenomenon is so named after Sir Robert Brown who discovered it in 1827.

Suspension and true solutions do not exhibit **Brownian movement**.

Electrical Double layer

The surface of colloidal particle acquires a positive charge by selective adsorption of a layer of positive ions around it. This layer attracts counterions from the medium which form a second layer of negative charges. The combination of the two layer of +ve and -ve charges around the sol particle was called **Helmholtz Double layer**. Helmholtz thought that positive charges next to the particle surface were fixed, while the layer of negative charges along with the medium were mobile.



■ **Figure 22.17**
Helmholtz double layer.

■ **Figure 22.18**
The electrical double layer (Stern).

More recent considerations have shown that the double layer is made of :

- a **Compact layer** of positive and negative charges which are fixed firmly on the particle surface.
- a **Diffuse layer** of counterions (negative ions) diffused into the medium containing positive ions.

(4) Coagulation or Precipitation

We know that the stability of a lyophobic sol is due to the adsorption of positive or negative ions by the dispersed particles. The repulsive forces between the charged particles do not allow them to settle. If, some how, the charge is removed, there is nothing to keep the particles apart from each other. They aggregate (or flocculate) and settle down under the action of gravity.

The flocculation and settling down of the discharged sol particles is called coagulation or precipitation of the sol.

How coagulation can be brought about?

The coagulation or precipitation of a given sol can be brought about in four ways :

- (a) By addition of electrolytes
- (b) By electrophoresis
- (c) By mixing two oppositely charged sols
- (d) By boiling

EMULSIONS

These are liquid-liquid colloidal systems. In other words, an emulsion may be defined as a dispersion of finely divided liquid droplets in another liquid.

Generally one of the two liquids is *water* and the other, which is immiscible with water, is designated as *oil*. Either liquid can constitute the dispersed phase.

Types of Emulsions

There are two types of emulsions.

(a) Oil-in-Water type (O/W type); (b) Water-in-Oil type (W/O type)

ADVANTAGES OF COLLOIDS



➤ Colloids allow the dispersion of normally insoluble materials, such as metallic gold or fats. These can then be used more easily, or absorbed more easily.

➤ Colloidal gold, for example, can be used in medicine to carry drugs and antibiotics, because it is highly non-reactive and non-toxic.

➤ Pharmaceutical industry makes use of colloidal solution preparation in many medicines. A wide variety of medicines are emulsions. An example is Cod Liver Oil.

➤ Paint industry also uses colloids in the preparation of paints.





➤ In milk, the colloidal suspension of the fats prevents the milk from being thick, and allows for easy absorption of the nutrients.

➤ Sewage water contains particles of dirt, mud etc. which are colloidal in nature and carry some electrical charge. These particles may be removed by using the phenomenon of electrophoresis.



➤ The sky is the empty space around earth and as such has no colour. It appears blue due to the scattering of light by the colloidal dust particles present in air (Tyndall effect).



➤ Asphalt emulsified in water and is used for building roads.



➤ The sugar present in milk produces lactic acid on fermentation. Ions produced by acid, destroy the charge on the colloidal particles present in milk, which then coagulate and separate as curd.



➤ Soap solution is colloidal in nature. It removes the dirt particles either by adsorption or by emulsifying the greasy matter sticking to the cloth.



➤ Large numbers of food particles which we use in our daily life are colloidal in nature.

Example: Milk, butter, & ice cream etc



